CPE 420: Intro to Data Mining

HomeWork 2

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AI Usage

Q1] Entropy and Gain Math Calculations were done by Copolite bing AI for faster results
Q2]Entropy and Gain Math Calculations were done by Copolite bing AI for faster results
Q3]Gini and Gain Math Calculations were done by Copolite bing AI for faster results
Q4]Math Calculations were done by Copolite bing AI for faster results

Problems

1. (3 points) Consider the given dataset. Employ Entropy and Error to find a split for attribute **Gender** in our attempt to build a binary decision classifier.

Customer Id	Gender	Marital Status	Home Type	Class
1	M	Married	Condo	Yes
2	M	Single	Apartment	Yes
3	M	Single	Apartment	Yes
4	M	Single	House	No
5	M	Single	Mansion	No
6	M	Single	Mansion	Yes
7	F	Single	Condo	No
8	F	Single	Condo	Yes
9	F	Single	Apartment	Yes
10	F	Divorced	House	No
11	M	Married	House	No
12	M	Married	Mansion	No
13	M	Married	Medium	Yes
14	M	Divorced	Mansion	No
15	F	Divorced	Condo	Yes
16	M	Divorced	Condo	No
17	F	Divorced	Apartment	No
18	F	Divorced	Apartment	No
19	M	Divorced	Apartment	Yes
20	F	Divorced	House	Yes

Before Gender:

P(yes) = 10 / 20 = 0.5 , =10/20 = 0.5
Entropy = − 0.5log0.5 −0.5log0.5 = 1
Error = 1−Max(0.5,0.5) = 1−0.5 = 0.5

After Gender:

	M (12)	F (8)
P(yes)	6/12 = 0.5	4/8 = 0.5
P(no)	6/12 = 0.5	4/8 = 0.5
Entropy	− 0.5log0.5 −0.5log0.5 = 1	− 0.5log0.5 −0.5log0.5 = 1
Error	1−0.5 = 0.5	1−0.5 = 0.5

Weighted Entropy = 12/20 x1 + 8/20 x 1 = 1

Weighted Error = 12/20 x 0.5 + 8/20 x 0.5 = 0.5

Gain Entropy = Entropy (before) − Weighted Entropy = 1 − 1 = 0

Error = Error (before) − Weighted Error = 0.5 − 0.5 = 0

Splitting on Gender doesnt improve for both entropy and Error

2. (5 points) Consider the training set given below for predicting lung cancer in patients based on their symptoms (chronic cough and weight loss) and other lifestyle and environmental attributes (tobacco smoking and exposure to radon). Draw a **two-level** decision tree obtained using **entropy** as the impurity measure. Show your steps clearly (i.e., the computation of information gain for every candidate attribute at the first and second levels of the decision tree must be shown). Compute the training error of the decision tree.

Tobacco Smoking	**Radon Exposure**	**Chronic Cough**	**Weight Loss**	**Lung Cancer (Class)**
Yes	Yes	Yes	No	Yes
Yes	No	No	No	No
Yes	No	Yes	No	Yes
Yes	No	Yes	Yes	Yes
No	No	Yes	No	No
Yes	No	Yes	No	Yes
No	Yes	No	Yes	Yes
No	No	Yes	Yes	No
No	No	Yes	No	No
No	No	No	Yes	No

Before:

P(yes) = 5 / 10 = 0.5 , =5 / 10 = 0.5

Entropy = − 0.5log0.5 −0.5log0.5 = 1

Root :

1.Tobacco Smoking

	Yes(5)	No(5)
P(yes)	4 / 5 = 0.8	1 / 5 = 0.2
P(no)	1 / 5 = 0.2	4 / 5 = 0.8
Entropy	− 0.8log0.8 −0.2log0.2 = 0.72188	− 0.2log0.2 −0.8log0.8 = 0.72188
Weighted Entropy = 5 / 10 x0.72188 + 5 / 10 x 0.72188 = 0.72188		
Gain Entropy = Entropy (before) − Weighted Entropy = 1 − 0.72188 = 0.27812		

2.Radon Exposure

	Yes(2)	No(8)
P(yes)	2 / 2 = 1	3 / 8 = 0.375
P(no)	0	5 / 8 = 0.625
Entropy	0	0.9544
Weighted Entropy = 2 / 10 x 0 + 8 / 10 x 0.9544 = 0.76352		
Gain Entropy = Entropy (before) − Weighted Entropy = 1 − 0.76352 = 0.23648		

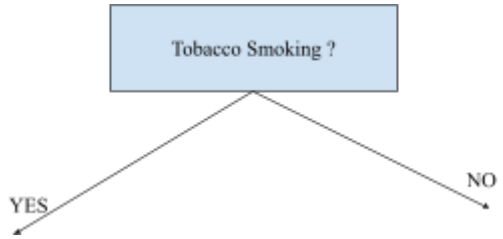
3.Chronic Cough

	Yes(7)	No(3)
P(yes)	4 / 7	1 / 3
P(no)	3 / 7	2 / 3
Entropy	0.9853	0.9183
Weighted Entropy = 7 / 10 x 0.9853 + 3 /10 x 0.9183 = 0.9652		
Gain Entropy = Entropy (before) − Weighted Entropy = 1 − 0.9652 = 0.0348		

4.WeightLoss

	Yes(4)	No(6)
P(yes)	2 / 4 = 2	3 / 6 = 3
P(no)	2 / 4 = 2	3 / 6 = 3
Entropy	1	1
Weighted Entropy = 4 / 10 + 6/10 = 1		
Gain Entropy = Entropy (before) − Weighted Entropy = 1 − 1 = 0		

Root choice would be Tobacco Smoking because it has the highest gain from all.



After choosing Root we work on second level Braches YES/NO as shown in the diagram above:

Yes Branch :

Tobacco Smoking	Radon Exposure	Chronic Cough	Weight Loss	Lung Cancer (Class)
Yes	Yes	Yes	No	Yes
Yes	No	No	No	No
Yes	No	Yes	No	Yes
Yes	No	Yes	Yes	Yes
No	No	Yes	No	No
Yes	No	Yes	No	Yes
No	Yes	No	Yes	Yes
No	No	Yes	Yes	No
No	No	Yes	No	No
No	No	No	Yes	No

1.Radon Exposure

	Yes(1)	No(4)
P(yes)	1 / 1 = 1	3 / 4
P(no)	0	1 / 4
Entropy	0	0.81125
Weighted Entropy = 1 / 5 x 0 + 4 /5 x 0.81125 = 0.649		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0.649 = 0.07288		

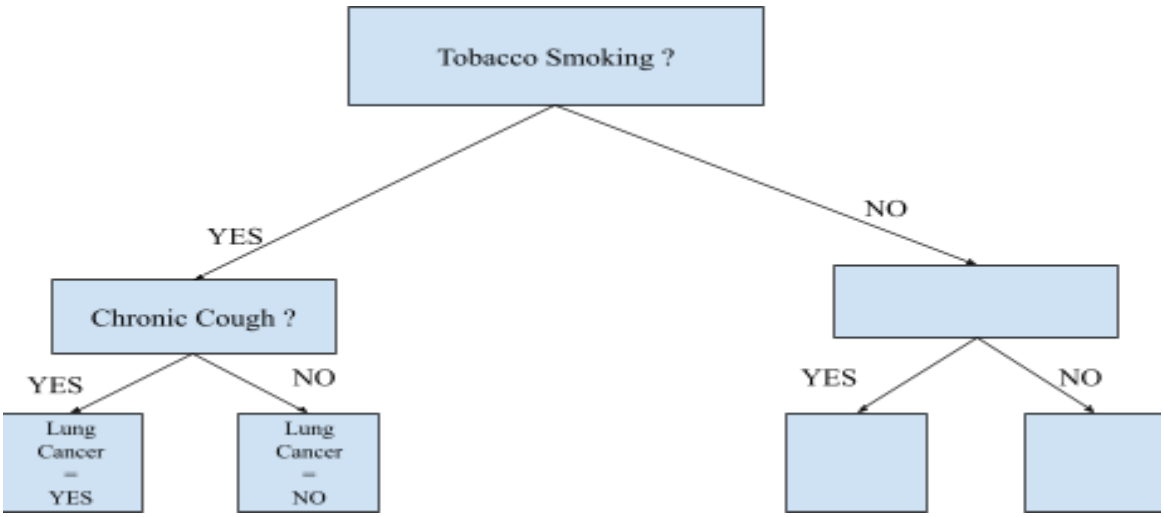
2.Chronic Cough

	Yes(4)	No(1)
P(yes)	4 / 4 = 1	1 / 1 = 1
P(no)	0	0
Entropy	0	0
Weighted Entropy = 0		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0 = 0.72188		

3.WeightLoss

	Yes(1)	No(4)
P(yes)	1 / 1 =1	3 / 4
P(no)	0	1 / 4
Entropy	0	0.81125
Weighted Entropy = 1 / 5 x 0 + 4 /5 x 0.81125 = 0.649		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0.649 = 0.07288		

Choice for Yes Branch is Chronic Cought as it has best gain



No Branch :

Tobacco Smoking	Radon Exposure	Chronic Cough	Weight Loss	Lung Cancer (Class)
Yes	Yes	Yes	No	Yes
Yes	No	No	No	No
Yes	No	Yes	No	Yes
Yes	No	Yes	Yes	Yes
No	No	Yes	No	No
Yes	No	Yes	No	Yes
No	Yes	No	Yes	Yes
No	No	Yes	Yes	No
No	No	Yes	No	No
No	No	No	Yes	No

1.Radon Exposure

	Yes(1)	No(4)
P(yes)	1 / 1 = 1	0
P(no)	0	4 / 4 = 1
Entropy	0	0
Weighted Entropy = 0		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0 = 0.72188		

2.Chronic Cough

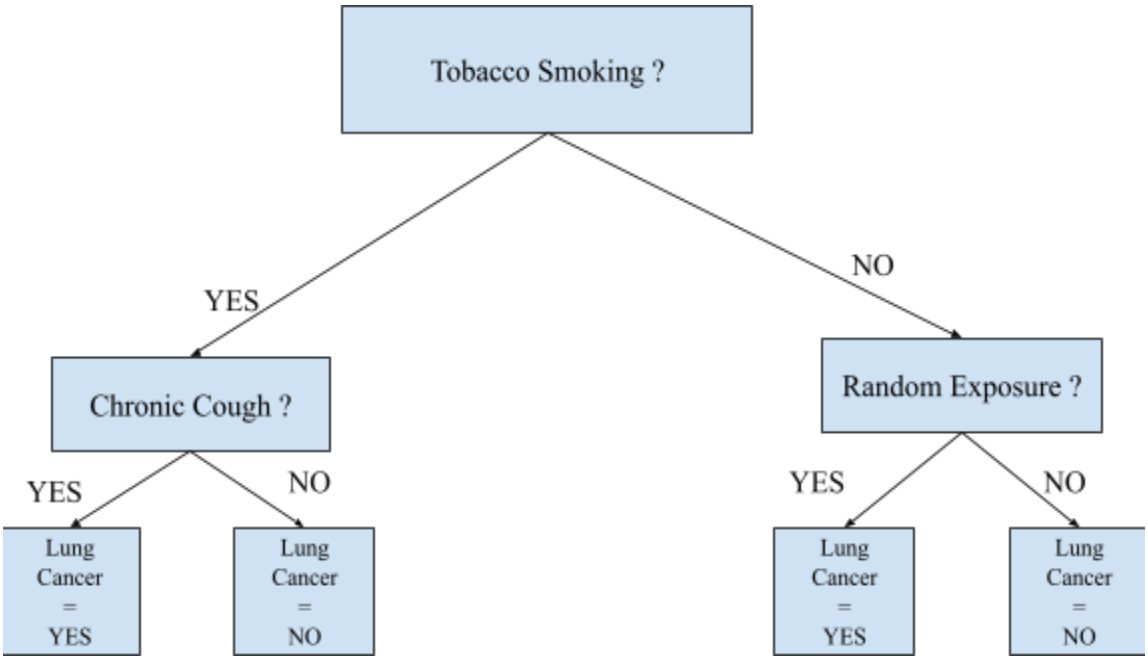
	Yes(3)	No(2)
P(yes)	0	1 / 2
P(no)	3 / 3 = 1	1 / 2
Entropy	0	1
Weighted Entropy = 3 /5 x 0 + 2 / 5 x 1 = 0.4		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0.4 = 0.32188		

3.WeightLoss

	Yes(3)	No(2)
P(yes)	1 / 3	0
P(no)	2 / 3	2 / 2 = 1
Entropy	0.9183	0
Weighted Entropy = 3 / 2 x 0.9183 + 0 = 0.55098		
Gain Entropy = Entropy (Tobacco) − Weighted Entropy = 0.72188 − 0.55098 =0.1709		

Choice for No Branch is Random Exposure as it has best gain

Decision Tree :



Training Error :

Tobacco Smoking	Radon Exposure	Chronic Cough	Weight Loss	Lung Cancer (Class)	Test ?
Yes	Yes	Yes	No	Yes	Pass
Yes	No	No	No	No	Pass
Yes	No	Yes	No	Yes	Pass
Yes	No	Yes	Yes	Yes	Pass
No	No	Yes	No	No	Pass
Yes	No	Yes	No	Yes	Pass
No	Yes	No	Yes	Yes	Pass
No	No	Yes	Yes	No	Pass
No	No	Yes	No	No	Pass
No	No	No	Yes	No	Pass

Training Error= 0 / 10 = 0

3. (6 points) Consider the following data set that contains 100 training examples (50 labeled as positive class while the remainder labeled as negative class).

X	Y	Z	No. of + Examples	No. of − Examples
1	1	1	5	0
1	1	0	0	20
1	0	1	20	0
1	0	0	0	5
0	1	1	10	0
0	1	0	15	0
0	0	1	0	10
0	0	0	0	15

(a) Build a two-level decision tree using **Gini index** as the criterion for splitting. You need to show your computations for each candidate splitting attribute at each level clearly to obtain full credit. What is the overall training error rate of the induced tree?
Note: we consider a tree with only 1 internal node and two leaf nodes as a one-level decision tree.

Before :

Total + = 5 + 20 + 10 + 15 = 50 , P+ = 50/100 = 0.5
Total − = 20 + 5 + 10 + 15 = 50 , P− = 0.5
Total Data = 100

Class Gini : 1 − ((0.5) ^ 2) − ((0.5) ^ 2) = 0.5

Root :

1. X

	1 (50)	0 (50)
P+	25 / 50= 0.5	25 / 50= 0.5
P−	25 / 50= 0.5	25 / 50= 0.5
Gini	1 − ((0.5) ^ 2) − ((0.5) ^ 2) = 0.5	1 − ((0.5) ^ 2) − ((0.5) ^ 2) = 0.5
Weighted Gini = 50 / 100 x (0.5) + 50 /100 x (0.5) = 0.5		
Gain Gini = Gini (Before) − Weighted Gini = 0.5 − 0.5 = 0		

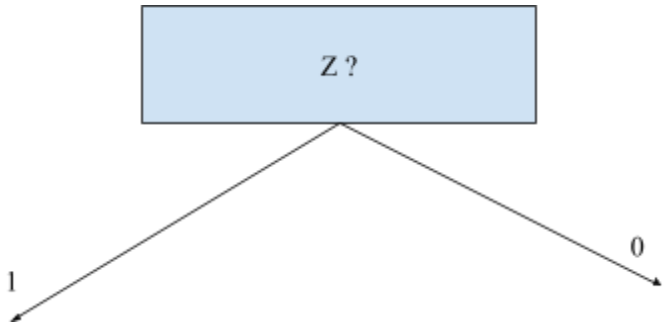
2. Y

	1 (50)	0 (50)
P+	30 / 50= 0.6	20 / 50= 0.4
P−	20 / 50= 0.4	30 / 50= 0.6
Gini	1 − ((0.6) ^ 2) − ((0.4) ^ 2) = 0.48	1 − ((0.4) ^ 2) − ((0.6) ^ 2) = 0.48
Weighted Gini = 50 / 100 x (0.48) + 50 /100 x (0.48) = 0.48		
Gain Gini = Gini (Before) − Weighted Gini = 0.5 − 0.48 = 0.02		

3. Z

	1 (45)	0 (55)
P+	35 / 45	15 / 55
P−	10 / 45	40 / 55
Gini	0.3457	0.3966
Weighted Gini = 45/ 100 x (0.3457) + 55/100 x (0.3966) = 0.3737		
Gain Gini = Gini (Before) − Weighted Gini = 0.5 − 0.3737 = 0.1263		

Choose Z as Root as it has the largest Gain



Level 2 Branches:

Z = 1 Branch :

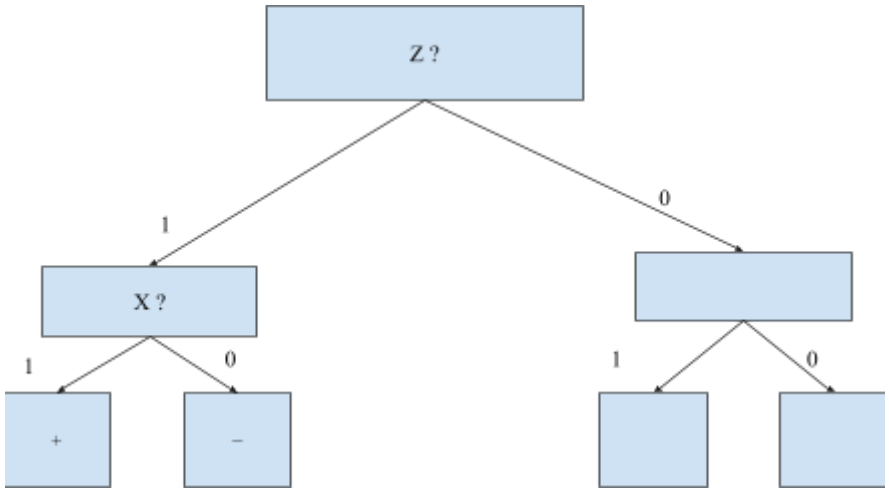
1. X

	1 (25)	0 (20)
P+	25 / 25 = 1	10 / 20
P−	0	10 / 20
Gini	0	0.5
Weighted Gini = 25 / 45 x (0) + 20 /45 x (0.5) = 0.2222		
Gain Gini = Gini (Root) − Weighted Gini = 0.3457 − 0.2222 = 0.1235		

2. Y

	1 (15)	0 (30)
P+	15 / 45	20 / 30
P−	0	10 / 30
Gini	0	0.4445
Weighted Gini = 15/ 45 x (0) + 30/45 x (0.4445) = 0.2963		
Gain Gini = Gini (Before) − Weighted Gini = 0.3457 − 0.2963 = 0.0494		

Choose X as 1 Branch as it has the largest Gain



Z = 0 Branch :

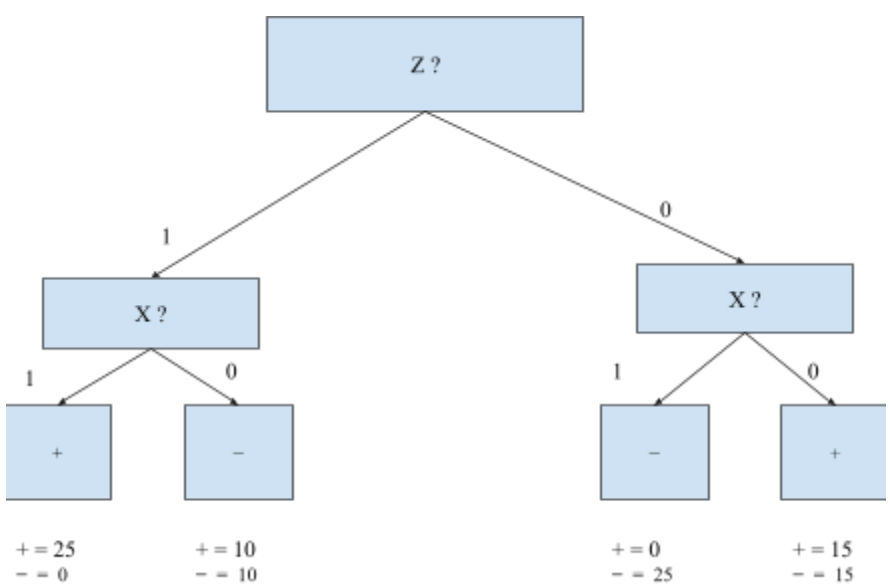
1. X

	1 (25)	0 (30)
P+	0	15 / 30
P−	25 / 25 = 1	15 / 30
Gini	0	0.5
Weighted Gini = 25 / 55x (0) + 30 /55 x (0.5) = 0.2727		
Gain Gini = Gini (Root) − Weighted Gini = 0.3966 − 0.2727 = 0.1239		

2. Y

	1 (35)	0 (20)
P+	15 / 35	0
P−	20 / 35	20 / 20 = 1
Gini	0.4898	0
Weighted Gini = 35/ 55 x (0.4898) + 20/55 x (0) = 0.3117		
Gain Gini = Gini (Before) − Weighted Gini = 0.3966 − 0.3117 = 0.0849		

Choose X as 0 Branch as it has the largest Gain



Training Error :

	Pass	Fall
Z1,X1(25)	25	0
Z1,X0 (20)	10	10
Z0,X1 (25)	25	0
Z0,X0 (30)	15	15

Training Error= 25 / 100 = 0.25

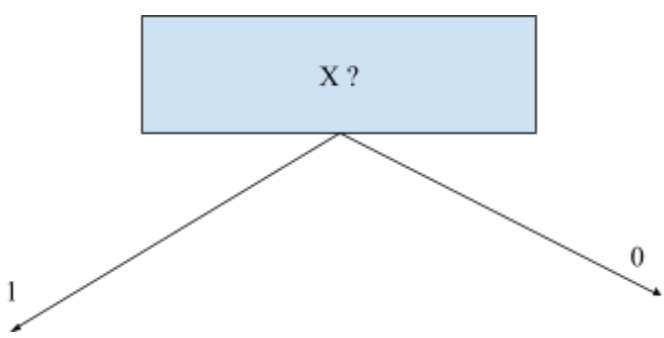
(b) Use variable **X** as the first splitting attribute, then choose the best available splitting attribute at each of the two successor nodes. What is the training error rate of the induced tree?

From Part A

Root :

1. X

	1 (50)	0 (50)
P+	25 / 50= 0.5	25 / 50= 0.5
P−	25 / 50= 0.5	25 / 50= 0.5
Gini	1 − ((0.5) ^ 2) − ((0.5) ^ 2) = 0.5	1 − ((0.5) ^ 2) − ((0.5) ^ 2) = 0.5
Weighted Gini = 50 / 100 x (0.5) + 50 /100 x (0.5) = 0.5		
Gain Gini = Gini (Before) − Weighted Gini = 0.5 − 0.5 = 0		



Level 2 Branches:

X= 1 Branch :

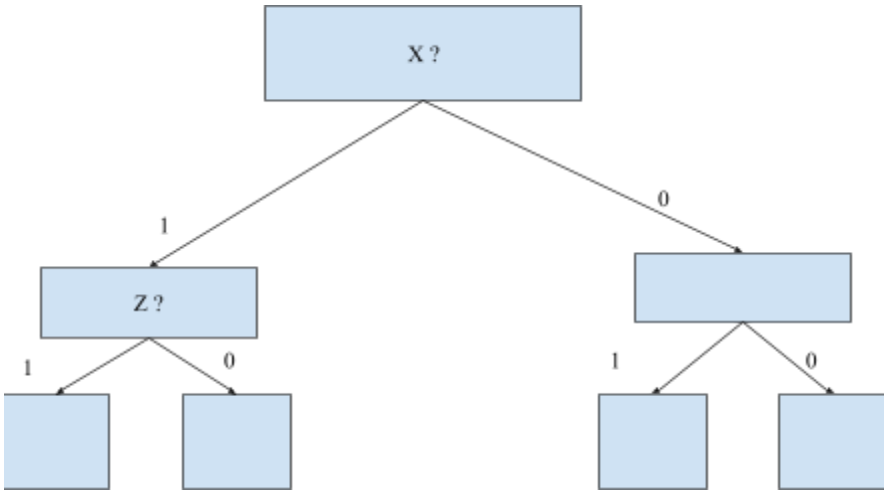
1.Y

	1 (25)	0 (25)
P+	5 / 25	20 / 25
P−	20 / 25	5 / 25
Gini	0.32	0.32
Weighted Gini = 0.32		
Gain Gini = Gini (Root) − Weighted Gini = 0.5 − 0.32 = 0.18		

2. Z

	1 (25)	0 (25)
P+	25 / 25 = 1	0
P−	0	25 / 25 = 1
Gini	0	0
Weighted Gini = 0		
Gain Gini = Gini (Root) − Weighted Gini = 0.5 − 0 = 0.5		

Choose Z as 1 Branch as it has the largest Gain



X= 0 Branch :

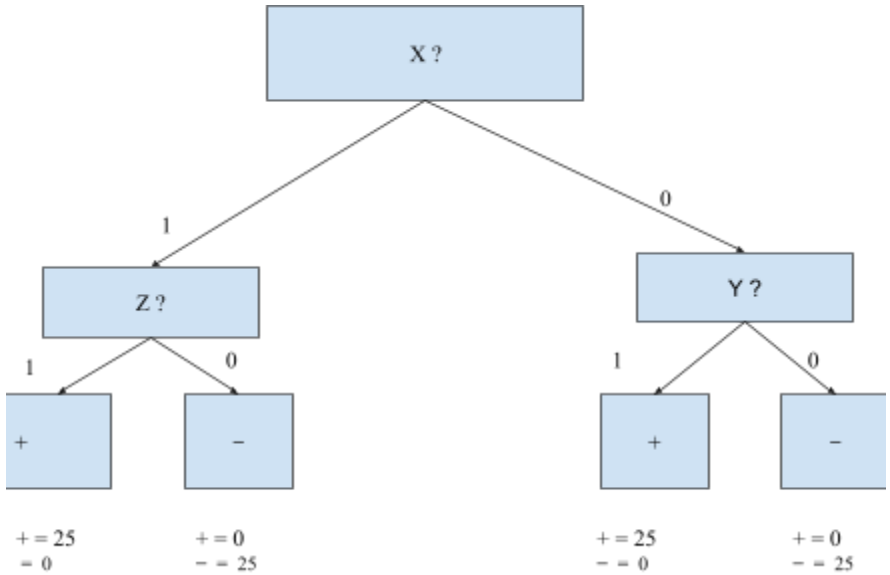
1.Y

	1 (25)	0 (25)
P+	25 / 25 = 1	0
P–	0	25 / 25 = 1
Gini	0	0
Weighted Gini = 0		
Gain Gini = Gini (Root) – Weighted Gini = 0.5 – 0 = 0.5		

2. Z

	1 (20)	0 (30)
P+	10 / 20	15 / 30
P–	10 / 20	15 / 30
Gini	0.5	0.5
Weighted Gini = 0.5		
Gain Gini = Gini (Root) – Weighted Gini = 0.5 – 0.5 = 0		

Choose Y as 0 Branch as it has the largest Gain



Training Error :

	Pass	Fall
X1,Z1(25)	25	0
X1,Z0 (25)	25	0
X0,Y1 (25)	25	0
X0,Y0 (25)	25	0

Training Error= 0

(c) Discuss the results obtained in parts (a) and (b) above. Comment on the suitability of the greedy heuristic used as the splitting attribute selection.

Part a :
The greedy chose Z as the root because its highest Gini gain (0.1263). However training error 0.25

Part b :
X is forced root with gain = 0. However training error 0 by using Z and Y at the second level.

Comment :
This shows that the greedy heuristic can be misleading. Because it picks the best split at each step without considering how future splits might work. Sometimes a split that looks bad at first like X enables perfect splits later, leading to a better tree.

At end while greedy are efficient, they are not guaranteed to find the optimal tree.

4. (6 points) Consider the table below.

Instance	a1	a2	a3	Class Label
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

(a) What is the entropy of this collection of training examples with respect to the positive class?

P(+) = 4 / 9 , P(-) =5 / 9
Entropy = − (4 / 9)log(4 / 9) −(5 / 9)log(5 / 9) = 0.9911

(b) What are the information gains of $a1$ and $a2$ relative to these training examples?

a1 :

T

P(+) = 3/4 , P(-) = 1 /4
Entropy = − (3/4)log(3/4) −(1 /4)log(1 /4) = 0.81125

F

P(+) = 1/5 , P(-) = 4 /5
Entropy = − (1/5)log(1/5) −(4 /5)log(4 /5) = 0.79638

Weighted entropy = 4/9(0.81125)+5/9(0.79638) = 0.80299
Gain = 0.9911 - 0.80299 = 0.18811

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a2 :

T

P(+) = 2/5 , P(-) = 3 /5
Entropy = − (2/5)log(2/5) −(3 /5)log(3 /5) = 0.97096

F

P(+) = 2/4 , P(-) = 2 /4
Entropy = − (2/4)log(2/4) −(2/4)log(2/4) = 1

Weighted entropy = 5/9(0.97096)+4/9(1) = 0.98386
Gain = 0.9911 - 0.98386 = 0.00724

(c) For $a3$, which is a continuous attribute, compute the information gain for every possible split.

Sorted a3 :
1 , 3 , 4 , 5 , 5 , 6 , 7 , 7 , 8
Midpoints :
2 , 3.5 , 4.5 , 5.5 , 6.5 , 7.5

Split 2	≤ 2.0 (1)	> 2.0 (8)
P+	1	3/8
P−	0	5 / 8
entropy	**0**	0.9544
Weighted entropy = 0.84836		
Gain entropy = entropy (Root) − Weighted entropy = 0.9911 − 0.84836 = 0.14274		

Split 3.5	≤ 3.5 (2)	> 3.5 (7)
P+	1/2	3/7
P−	1/2	4 / 7
entropy	1	0.9853
Weighted entropy = 0.9885		
Gain entropy = entropy (Root) − Weighted entropy = 0.9911 − 0.9885 = 0.0026		

Split 4.5	≤ 4.5 (3)	> 4.5 (6)
P+	2/3	2/6
P−	1/3	4 / 6
entropy	0.9183	0.9183
Weighted entropy = 0.9183		
Gain entropy = entropy (Root) − Weighted entropy = 0.0728		

Split 5.5	≤ 4.5 (5)	> 4.5 (4)
P+	2/5	2/4
P−	3 / 5	2/4
entropy	0.97096	1
Weighted entropy = 0.98386		
Gain entropy = entropy (Root) − Weighted entropy = 0.00724		

Split 6.5	≤ 6.5 (5)	> 6.5 (3)
P+	3/6	1/3
P−	3/6	2/3
entropy	1	0.9183
Weighted entropy = 0.9728		
Gain entropy = entropy (Root) − Weighted entropy = 0.0183		

Split 7.5	≤ 7.5	> 7.5
P+	4 / 8	0
P−	4 / 8	1
entropy	1	0
Weighted entropy = 0.8889		
Gain entropy = entropy (Root) − Weighted entropy = 0.1022		

(d) What is the best split (among $a1$, $a2$, and $a3$) according to the information gain?

The best split is a1 = 0.1881.

(e) What is the best split (between $a1$ and $a2$) according to the classification error rate?

a1 :

T

P(+) = 3/4 , P(-) = 1 /4
Error = 1 - 3/4 = 0.25

F

P(+) = 1/5 , P(-) = 4 /5
Error = 1 - 4 /5 = 0.2

Wighted Error = 4/9 (0.25) + 5 / 9 (0.2) = 0.2222
Error = 0.4444−0.2222=0.2222

========== ========== ========== ==========

a2 :

T

P(+) = 2/5 , P(-) = 3 /5
Error = 1 - 3/4 = 0.4

F

P(+) = 2/4 , P(-) = 2 /4
Error = 1 - 2/4 = 0.5

Wighted Error = 5 /9 (0.4) + 4 / 9 (0.5) = 0.4444
Error =0.4444−0.4444=0

a1 is best split

(f) What is the best split (between $a1$ and $a2$) according to the Gini index?

Gini Root =1−0.1975−0.3086=0.4939

a1 :

T

P(+) = 3/4 , P(-) = 1 /4

Gini= 0.375

F

P(+) = 1/5 , P(-) = 4 /5

Gini= 0.32

Wighted Gini= 0.34445

gain= 0.4939−0.34445=0.14945

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a2 :

T

P(+) = 2/5 , P(-) = 3 /5

Gini= 0.48

F

P(+) = 2/4 , P(-) = 2 /4

Gini= 0.5

Wighted Gini=0.48889

gain=0.4939−0.48889=0.00501

a1 is best split